Model Preparation and Training

After completing the exploratory analysis, the next step of the project focused on preparing the dataset for predictive modeling and training machine learning models capable of estimating track popularity based on audio features and musical attributes. The original dataset contained 232,725 observations and 15 variables after removing non-informative identifiers such as track_name, artist_name, and track_id. Since several of the remaining variables were categorical, an encoding step was required before training the models. To make the categorical variables usable for machine learning algorithms, one-hot encoding was applied to the variables genre, key, mode, and time_signature. This process converts each category into a binary feature representing the presence or absence of that specific category. As a result, the number of variables in the dataset increased from 15 to 57 columns, reflecting the expanded set of encoded features. The final dataset therefore contained both the original numerical audio descriptors and the newly created binary indicators representing the different musical categories. Once the encoding step was completed, the dataset was divided into training and test sets in order to evaluate model performance on unseen data. The training set contains 186,180 observations, while the test set contains 46,545 observations, corresponding to approximately 80% and 20% of the data, respectively. The feature matrix used for training therefore includes 56 predictor variables, while the target variable is the popularity score, which ranges from 0 to 100.

Random Forest Model

To model the relationship between musical features and track popularity, a Random Forest Regressor was trained. Random Forest is a tree-based ensemble algorithm that combines multiple decision trees in order to capture complex and non-linear relationships between variables. This type of model is particularly suitable for datasets like this one because it can handle high-dimensional feature spaces and is relatively robust to outliers and non-linear interactions. The model was trained using the training dataset and evaluated on both the training and test sets using several performance metrics, including Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and the coefficient of determination (R^2).

The results obtained are the following:

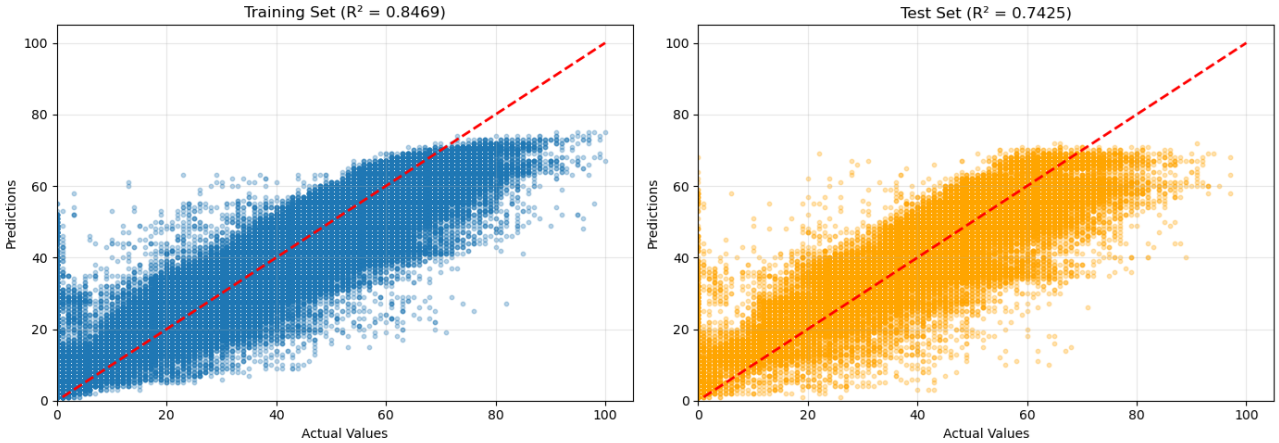
Metric	Training	Test
RMSE	7.11	9.26
MAE	5.26	7.02
R^2	0.8468	0.7425

These results indicate that the Random Forest model is able to explain approximately 74% of the variance in track popularity on the test dataset, which suggests a strong predictive capability. However, the difference between training and test performance suggests the presence of moderate overfitting. The test-to-train RMSE ratio is approximately 1.30, which indicates that the model performs significantly better on the training data than on unseen observations. This behavior is typical of flexible models such as Random Forests, which can capture complex patterns in the training data but may partially adapt to noise or dataset-specific structures.

Model Predictions

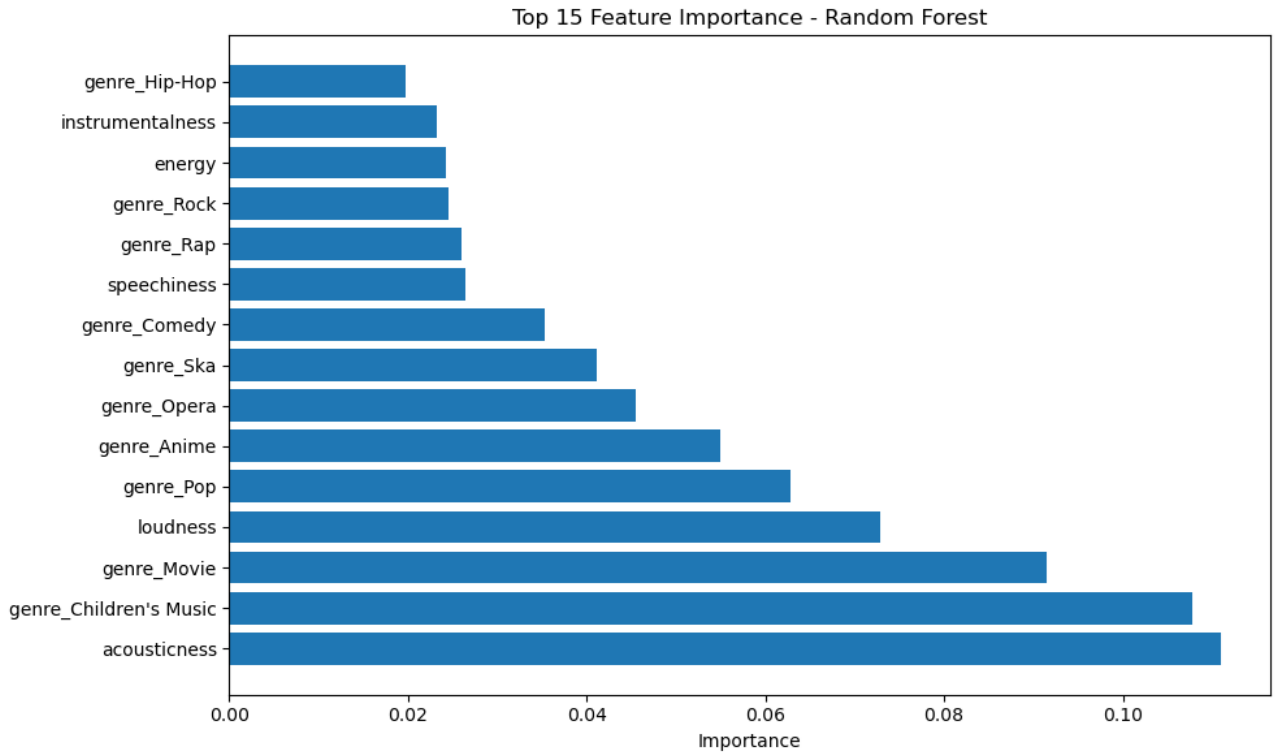
To better understand the model's behavior, the predicted popularity values were compared to the actual popularity scores. The following visualization shows the relationship between real values and predicted values for both the training and test datasets. Ideally, predictions should lie close to the diagonal line, which represents perfect predictions. In the training dataset, the points are relatively concentrated around this line, reflecting the model's strong fit to the training data. In the test dataset, the dispersion of points is slightly larger, which is consistent with the previously observed overfitting. Nevertheless, the predictions still follow the overall structure of the data, indicating that

the model is able to capture a substantial portion of the underlying relationships.



Feature Importance Analysis

One advantage of tree-based models such as Random Forest is the ability to estimate feature importance, which measures how much each variable contributes to the predictive performance of the model. The analysis of feature importance highlights several interesting results. The most important variable in the model is acousticness, followed by several genre-related features such as Children's Music, Movie, and Pop. Among the numerical audio features, acousticness, loudness, speechiness, energy, and instrumentalness also play a significant role in predicting track popularity.



These findings suggest that both musical style (genre) and audio characteristics contribute to explaining the popularity of songs. In particular, the prominence of genre-related features reinforces the earlier findings from the exploratory analysis, which indicated that genre has a strong association with popularity.

Baseline Model: Linear Regression

To better evaluate the effectiveness of the Random Forest model, a Linear Regression model was also trained as a baseline. Linear regression assumes a linear relationship between the input features

and the target variable, making it a simpler and more interpretable model compared to tree-based methods.

The performance of the linear regression model is summarized below:

Metric	Training	Test
RMSE	9.59	9.61
MAE	7.25	7.27
R^2	0.7218	0.7228

The results show that the linear model explains approximately 72% of the variance in track popularity on the test dataset.

Model Comparison

Comparing the two models reveals that the Random Forest model outperforms the linear regression baseline, achieving lower prediction errors and a higher R^2 score. However, the improvement is relatively moderate rather than dramatic. While Random Forest achieves a better fit to the training data, the difference in test performance compared to linear regression is not extremely large. This observation suggests that although the relationship between audio features and popularity is not strictly linear, a significant portion of the predictive signal can still be captured by simpler models. At the same time, the presence of moderate overfitting in the Random Forest model indicates that additional tuning or regularization techniques (such as limiting tree depth or applying alternative ensemble methods like Gradient Boosting or XGBoost) could potentially improve generalization performance.